

Ergodic Secrecy Capacity Analysis of Reconfigurable Intelligent Surface-Assisted Wireless Networks with SC and MRC Diversity Schemes

Abstract—This paper discusses the ergodic secrecy capacity (ESC) of a reconfigurable intelligent surface (RIS)-assisted wireless communication system, using Selection Combining (SC) and Maximal Ratio Combining (MRC) diversity schemes with a passive multi-antenna eavesdropper. A single-antenna source communicates with a multi-antenna legitimate destination, while the eavesdropper tries to decode the message. All communication takes place through an RIS comprising N passive reflecting elements, while the direct links from the source to the destination and eavesdropper are blocked by obstacles. The closed-form expressions for the ESC are derived under both SC and MRC diversity schemes, and the analytical results are validated through extensive Monte Carlo simulations. The influence of key system parameters, including the number of RIS reflecting elements, the antenna configurations at the destination and eavesdropper, and the average eavesdropper SNR, is thoroughly examined, showing that the ESC improves with a larger number of RIS reflecting elements and destination antennas, whereas it decreases with an increasing number of eavesdropper antennas and average eavesdropper SNR. A comparative numerical evaluation also shows that the MRC scheme can achieve higher secrecy capacity than the SC scheme by combining the signals collected across different antenna branches. These findings confirm that the combination of RIS-aided propagation and receiver diversity is an effective and practical solution to improve wireless security in next-generation wireless networks.

Index Terms—Reconfigurable intelligent surface (RIS), physical layer security (PLS), ergodic secrecy capacity (ESC), selection combining (SC), maximal ratio combining (MRC).

I. INTRODUCTION

Beyond fifth-generation (5G) and sixth-generation (6G) wireless communication systems are rapidly evolving, and the demand for secure, reliable, and energy-efficient wireless transmission has grown significantly. Recently, Reconfigurable Intelligent Surfaces (RISs) have been proposed as a new technology with low power consumption and low hardware complexity, which can intelligently reconfigure the wireless propagation environment by utilizing the programmable passive reflecting elements to improve signal quality, coverage, and spectral efficiency [1]. Moreover, the combination of RIS and physical layer security (PLS) is an effective solution for enhancing wireless communication security by strengthening the legitimate link while minimizing information leakage to the eavesdropper [2].

A. Related Works

Recently, Khoshafa et al. [3] studied the physical layer security of RIS-aided cognitive radio networks (CRNs) and

provided closed-form expressions for the secrecy outage probability (SOP) and probability of non-zero secrecy capacity (PNZSC). However, their study mainly focused on outage-related secrecy measures and did not address the ergodic secrecy capacity (ESC). The secrecy performance of RIS-assisted RF-underwater optical wireless communication (RF-UOWC) systems was explored by Hossain et al. [4], who derived analytical expressions for the ESC, SOP, and effective secrecy throughput. However, the proposed framework was developed for mixed RF-UOWC networks rather than conventional terrestrial RIS-assisted systems. Shahriyer et al. [5] discussed the ergodic secrecy capacity of an opportunistic relay-assisted multicast transmission system over generalized shadowed fading channels. Unlike RIS-assisted communications, their work was based on active relay selection to enhance secrecy performance. In [6], Abbasi Mosleh et al. provided exact closed-form and asymptotic expressions for the ergodic capacity of RIS-assisted MIMO systems in Rayleigh-Rician fading channels. While their analytical approach offers a valuable capacity perspective, secrecy performance was not considered. Yang et al. [7] investigated RIS-assisted wireless-powered relay systems and developed analytical models of ergodic capacity for various transmission modes. Their contribution was focused on capacity enhancement rather than ensuring secure communication. Hung et al. [8] studied the ergodic rate and effective capacity of RIS-assisted NOMA networks over Nakagami-m fading channels. This study focused mainly on communication efficiency, rather than measuring secrecy performance metrics. Yadav et al. [9] derived the exact and asymptotic secrecy outage probability and ESC of STAR-RIS-assisted wireless communication. They did not include other receiver diversity techniques, such as SC and MRC. Zhou et al. [10] explored the secrecy outage probability of an RIS-assisted satellite-UAV relay channel under MRC eavesdropping. They investigated only outage-based secrecy analysis instead of average secrecy capacity. Xu et al. [11] studied the ergodic secrecy rate of RIS-assisted communication systems, where the phase shifts of RIS are discrete and multiple eavesdroppers exist. The impact of practical receiver diversity combining on secrecy performance was not considered. Cao et al. [12] studied the ergodic secrecy capacity of RIS-assisted wireless-powered communication systems, and provided exact and asymptotic closed-form expressions for various deployment scenarios. However, a comparative study based on SC and MRC diversity reception was not included in their study.

TABLE I
COMPARISON WITH EXISTING WORKS

Ref.	Channel	RIS	SC	MRC	Security	Multi-Antenna	Metrics
[3]	Rayleigh	✓	✓	✓	✓	×	SOP, PNZSC
[4]	Mixed RF-UOWC	✓	×	×	✓	×	ASC, SOP, SPSC, EST
[5]	κ - μ Shadowed	×	×	×	✓	✓	PNSMC, SOPM, ESMC
[6]	Rayleigh-Rician	✓	×	×	×	✓	EC
[7]	Rician	✓	×	×	×	✓	EC
[8]	Nakagami- m	✓	×	×	×	×	ER, EC
[9]	Nakagami- m	✓	×	×	✓	✓	SOP, ESC
[10]	Shadowed-Rician	✓	×	✓	✓	×	SOP
[11]	Rayleigh	✓	×	×	✓	×	ESR
[12]	Rayleigh	✓	×	×	✓	×	ESC
This Work	Rayleigh	✓	✓	✓	✓	✓	ESC

Although communication using RIS is a growing research interest, the vast majority of works in the literature have concentrated on the ergodic capacity, ergodic secrecy rate, or secrecy outage probability, typically in an application-specific scenario. The ergodic secrecy capacity of RIS-assisted networks under practical receiver diversity remains largely unexplored, and a direct comparison between Selection Combining (SC) and Maximal Ratio Combining (MRC) has not been reported. To counter this deficiency, this paper develops the exact closed-form expressions of ESC for RIS-aided networks under both SC and MRC, and provides a detailed comparative performance analysis.

B. Contributions

To address this gap in the existing literature, an analytical framework is developed for the ergodic secrecy capacity (ESC) of RIS-enabled wireless networks with both SC and MRC diversity schemes. The significant contributions of the presented work are briefly summarized as follows:

- **First**, exact closed-form expressions for the ergodic secrecy capacity (ESC) of RIS-assisted wireless networks with SC and MRC diversity schemes are derived.
- **Second**, a detailed performance comparison between the SC and MRC diversity schemes is conducted to assess the effect on the ESC of RIS-assisted wireless networks.
- **Third**, the impact of the number of RIS reflecting elements, the average SNR at the eavesdropper, and the antenna configurations at the legitimate receiver and the eavesdropper on the ESC performance is thoroughly investigated for both SC and MRC diversity schemes.
- **Finally**, the analytical results are checked using extensive Monte Carlo simulations, showing good agreement between the analytical results and the simulation results.

The remaining part of the paper is arranged as follows. The system structure and channel assumptions are presented in Section II. Section III develops the analytical expressions for the secrecy metrics. Section IV provides numerical evaluations along with simulation validation, and Section V concludes the study.

II. SYSTEM MODEL

A. Network Configuration and Channel Environment

We consider an RIS-assisted secure communication system consisting of a single-antenna source S , a legitimate des-

tinuation D with N_d antennas and an eavesdropper E with N_e antennas. There is a physical barrier between S and D and between S and E . Thus, an RIS R , consisting of N passive reflecting elements, is used to provide an alternate propagation route, as shown in Fig. 1. The source-to-RIS, RIS-to-destination, and RIS-to-eavesdropper links are assumed to have independent Rayleigh fading under a block-fading channel model.

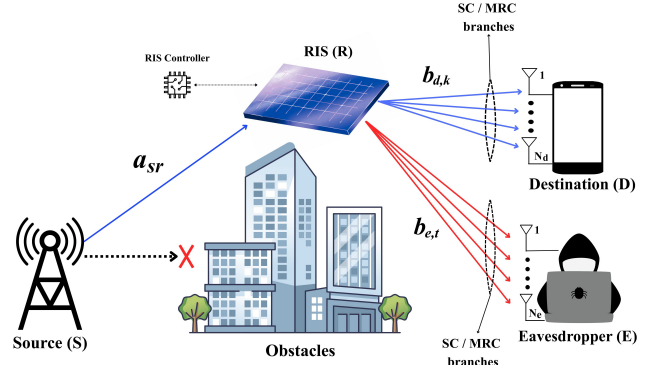


Fig. 1. RIS-assisted secure system with SC and MRC diversity.

B. Signal and Channel Modeling

Let x denote the transmitted information symbol that satisfies $\mathbb{E}[|x|^2] = 1$, and P_s represent the transmit power. The source-RIS channel is denoted by $\mathbf{a}_{sr} \in \mathbb{C}^{N \times 1}$. The vectors $\mathbf{b}_{d,k} \in \mathbb{C}^{N \times 1}$ and $\mathbf{b}_{e,t} \in \mathbb{C}^{N \times 1}$ represent the channel vectors from RIS-to-destination and RIS-to-eavesdropper with the k -th and t -th receiver branches, respectively. The RIS phase response is represented by

$$\Psi = \text{diag}(e^{j\psi_1}, e^{j\psi_2}, \dots, e^{j\psi_N}), \quad (1)$$

where ψ_n denotes the phase change due to the n -th reflecting element.

The received signal at the k -th antenna of the destination is provided by

$$r_{d,k} = \sqrt{P_s} \mathbf{b}_{d,k}^H \Psi \mathbf{a}_{sr} x + n_{d,k}, \quad k = 1, \dots, N_d, \quad (2)$$

while the signal received at the t -th antenna of the eavesdropper is expressed as

$$r_{e,t} = \sqrt{P_s} \mathbf{b}_{e,t}^H \Psi \mathbf{a}_{sr} x + n_{e,t}, \quad t = 1, \dots, N_e. \quad (3)$$

Here, $r_{d,k}$ and $r_{e,t}$ denote the received signals at the k -th destination antenna and t -th eavesdropper antenna, respectively. The terms $n_{d,k}$ and $n_{e,t}$ are independent zero-mean AWGN with variance σ_n^2 .

C. Receiver Diversity

Both Selection Combining (SC) and Maximal Ratio Combining (MRC) are considered at the destination and eavesdropper. Under SC, the receiver chooses the branch with the highest instantaneous SNR. Thus, the output SNRs are

$$\eta_d^{\text{SC}} = \frac{P_s}{\sigma_n^2} \max_k |\mathbf{b}_{d,k}^H \Psi \mathbf{a}_{sr}|^2, \quad \eta_e^{\text{SC}} = \frac{P_s}{\sigma_n^2} \max_t |\mathbf{b}_{e,t}^H \Psi \mathbf{a}_{sr}|^2. \quad (4)$$

For MRC, the signals from all available branches are combined together. The corresponding output SNRs are given by

$$\eta_d^{\text{MRC}} = \sum_{k=1}^{N_d} \frac{P_s}{\sigma_n^2} |\mathbf{b}_{d,k}^H \Psi \mathbf{a}_{sr}|^2, \quad \eta_e^{\text{MRC}} = \sum_{t=1}^{N_e} \frac{P_s}{\sigma_n^2} |\mathbf{b}_{e,t}^H \Psi \mathbf{a}_{sr}|^2. \quad (5)$$

The resulting output SNRs are subsequently used to characterize the secrecy performance of the RIS-assisted system considered under the SC and MRC diversity schemes.

III. PROBLEM FORMULATION

A. PDFs of the Destination and Eavesdropper SNRs for RIS-Aided Networks under SC Diversity

The PDFs of the destination and the eavesdropper SNR are presented in (6) and (7), respectively. Here, γ_d and γ_e represent the corresponding instantaneous SNRs, while ω_d and ω_e are their average values. The parameters Λ_d and Λ_e characterize the exponential interference-plus-noise distributions. The summation indices k and t refer to the diversity branches, with the associated terms involving $\binom{N_d-1}{k}$ and $\binom{N_e-1}{t}$. The PDFs of the received SNRs at the destination are provided by [3], eq. (18)

$$f_{\gamma_d}^{\text{SC}}(\gamma_d) = \sum_{k=0}^{N_d-1} A_k \frac{1 + B_d + C_k \gamma_d}{(B_d + C_k \gamma_d)^2} e^{-C_k \gamma_d}. \quad (6)$$

where

$$A_k = \frac{N_d(-1)^k}{\omega_d \Lambda_d} \binom{N_d-1}{k}, \quad B_d = \frac{1}{\Lambda_d}, \quad C_k = \frac{k+1}{\omega_d}.$$

The PDFs of the received SNRs at the eavesdropper are given by [3], eq. (18)

$$f_{\gamma_e}^{\text{SC}}(\gamma_e) = \sum_{t=0}^{N_e-1} A_t \frac{1 + B_e + C_t \gamma_e}{(B_e + C_t \gamma_e)^2} e^{-C_t \gamma_e}. \quad (7)$$

where

$$A_t = \frac{N_e(-1)^t}{\omega_e \Lambda_e} \binom{N_e-1}{t}, \quad B_e = \frac{1}{\Lambda_e}, \quad C_t = \frac{t+1}{\omega_e}.$$

B. Ergodic Secrecy Capacity for SC Diversity

The Ergodic Secrecy Capacity (ESC) is expressed as

$$\begin{aligned} \langle C_s^{\text{SC}} \rangle &= \int_0^\infty \log_e(1 + \gamma_d) f_{\gamma_d}^{\text{SC}}(\gamma_d) d\gamma_d \\ &\quad - \int_0^\infty \log_e(1 + \gamma_e) f_{\gamma_e}^{\text{SC}}(\gamma_e) d\gamma_e, \quad \gamma_d > \gamma_e. \end{aligned} \quad (8)$$

Using the identities from [13] (3.352.4), the closed-form equation of ESC can be written as

$$\begin{aligned} \langle C_s^{\text{SC}} \rangle &= \sum_{k=0}^{N_d-1} \frac{A_k}{C_k(B_d - C_k)} [e^{B_d} \text{Ei}(-B_d) - e^{C_k} \text{Ei}(-C_k)] \\ &\quad - \sum_{t=0}^{N_e-1} \frac{A_t}{C_t(B_e - C_t)} [e^{B_e} \text{Ei}(-B_e) - e^{C_t} \text{Ei}(-C_t)]. \end{aligned} \quad (9)$$

C. PDFs of the Destination and Eavesdropper SNRs for RIS-Aided Networks under MRC Diversity

The probability density functions (PDFs) of the received SNRs at the legitimate receiver and the eavesdropper are presented in (10) and (11), respectively [3], eq. (20).

$$f_{\gamma_d}^{\text{MRC}}(\gamma) = \sum_{k=0}^{N_d} A_{d,k} \frac{\gamma^{N_d-1} e^{-C_d \gamma}}{(B_d + C_d \gamma)^{k+1}} \quad (10)$$

$$f_{\gamma_e}^{\text{MRC}}(\gamma) = \sum_{t=0}^{N_e} A_{e,t} \frac{\gamma^{N_e-1} e^{-C_e \gamma}}{(B_e + C_e \gamma)^{t+1}} \quad (11)$$

where

$$\begin{aligned} A_{d,k} &= \frac{\Gamma(k+1) \binom{N_d}{k}}{\Gamma(N_d) \Lambda_d \omega_d^{N_d}}, & B_d &= \frac{1}{\Lambda_d}, & C_d &= \frac{1}{\omega_d}. \\ A_{e,t} &= \frac{\Gamma(t+1) \binom{N_e}{t}}{\Gamma(N_e) \Lambda_e \omega_e^{N_e}}, & B_e &= \frac{1}{\Lambda_e}, & C_e &= \frac{1}{\omega_e}. \end{aligned}$$

D. Ergodic Secrecy Capacity for MRC Diversity

$$\begin{aligned} \langle C_s^{\text{MRC}} \rangle &= \int_0^\infty \log_e(1 + \gamma_d) f_{\gamma_d}^{\text{MRC}}(\gamma_d) d\gamma_d \\ &\quad - \int_0^\infty \log_e(1 + \gamma_e) f_{\gamma_e}^{\text{MRC}}(\gamma_e) d\gamma_e, \quad \gamma_d > \gamma_e. \end{aligned} \quad (12)$$

The exact equation of ESC under the MRC diversity scheme is obtained as

$$\begin{aligned} \langle C_s^{\text{MRC}} \rangle &= \sum_{k=0}^{N_d} A_{d,k} C_d^{-(k+1)} \sum_{i=0}^{N_d-1} \binom{N_d-1}{i} (-\beta_d)^{N_d-1-i} \mathcal{I}_{d,p} \\ &\quad - \sum_{t=0}^{N_e} A_{e,t} C_e^{-(t+1)} \sum_{i=0}^{N_e-1} \binom{N_e-1}{i} (-\beta_e)^{N_e-1-i} \mathcal{I}_{e,q}, \end{aligned} \quad (13)$$

where

$$p = i - k - 1, \quad q = i - t - 1.$$

The required auxiliary integral is defined as

$$\mathcal{I}_{i,n} = \int_0^\infty (\gamma + \beta_i)^n e^{-C_i \gamma} \ln(1 + \gamma) d\gamma, \quad i \in \{d, e\}, \quad (14)$$

where

$$\beta_i = \frac{B_i}{C_i}.$$

For $n \geq 0$, using [13], Eqs. 2.321.2, 3.381.3] and $\Gamma(0, x) = -\text{Ei}(-x)$ [13], Eq. 8.359.1], the closed-form solution is given by

$$\mathcal{I}_{i,n} = e^{C_i} \sum_{r=0}^n \frac{n!}{r! C_i^{n-r+1}} \sum_{\ell=0}^r \binom{r}{\ell} (\beta_i - 1)^{r-\ell} \frac{\Gamma(\ell, C_i)}{C_i^\ell}. \quad (15)$$

For $n = -1$,

$$\mathcal{I}_{i,-1} = -e^{C_i \beta_i} \int_{C_i}^\infty \frac{e^{-w(\beta_i-1)} \text{Ei}(-w)}{w} dw. \quad (16)$$

The rest of the negative order terms are found recursively as

$$\mathcal{I}_{i,n-1} = \frac{1}{n} \frac{\partial \mathcal{I}_{i,n}}{\partial \beta_i}, \quad n \neq 0. \quad (17)$$

By substituting these terms into (13), we obtain the closed-form ESC equation for MRC diversity scheme.

IV. NUMERICAL RESULTS

The analytical expressions for ESC, obtained in Section III, for both SC and MRC, are numerically evaluated using MATLAB. The analytical results are then compared with Monte Carlo simulation results to verify the final equations.

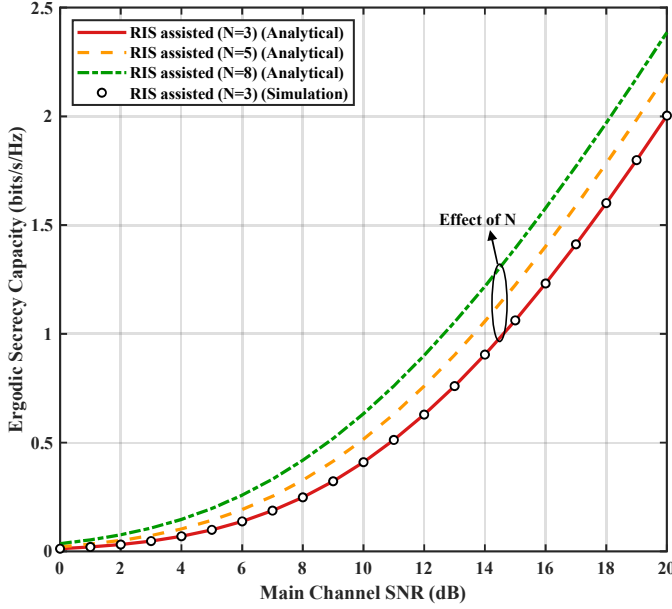


Fig. 2. Effect of N on ESC under SC for $N_d = 2$, $N_e = 5$, and $\bar{\gamma}_e = 10$ dB.

Figure 2 presents the ESC performance for different numbers of RIS reflecting elements, $N = 3, 5$, and 8 , under the SC diversity scheme. The ESC monotonically increases with the SNR of the main channel, suggesting improved secrecy performance at higher SNR values for all considered configurations. The more RIS reflecting elements, the more noticeable the performance gain, due to a better effective received signal strength. Among the considered cases, $N = 8$ achieves the highest ESC, whereas $N = 3$ yields the lowest ESC. The good match between the analytical and simulation results also confirms the validity of the developed analytical model.

Figure 3 illustrates the impact of the average eavesdropper SNR on the ESC of the considered system with SC. The results are presented for $\bar{\gamma}_e = 5, 10$, and 15 dB. It is observed that as the eavesdropper average SNR ($\bar{\gamma}_e$) increases, the ESC decreases, since a stronger eavesdropper link enhances its interception capability and degrades the overall secrecy performance. In the considered cases, the highest secrecy capacity is obtained at $\bar{\gamma}_e = 5$ dB while the lowest is obtained

at $\bar{\gamma}_e = 15$ dB. Therefore, reducing the average eavesdropper SNR improves secrecy performance by limiting the eavesdropper's ability to intercept the transmitted information.

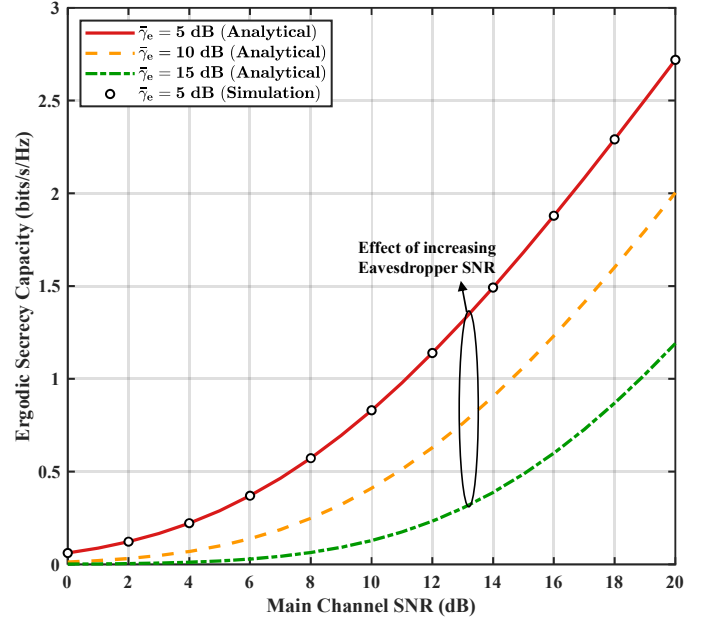


Fig. 3. Impact of $\bar{\gamma}_e$ on ESC under SC for $N_d = 2$, $N_e = 5$, and $N = 3$.

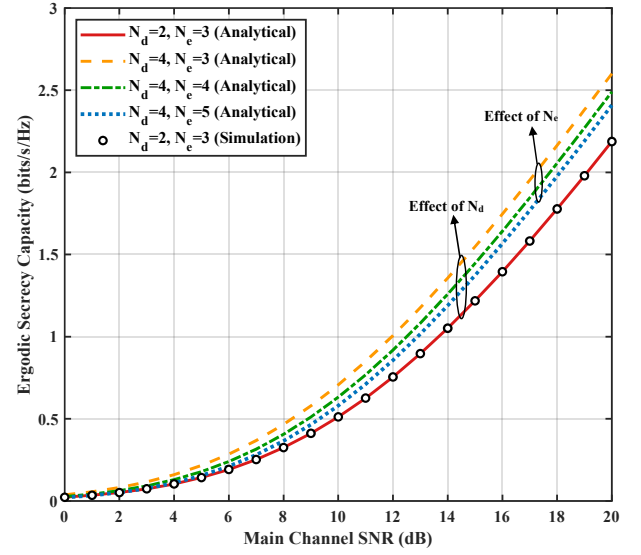


Fig. 4. Variation of ESC with respect to N_d and N_e under SC for $N = 3$ and $\bar{\gamma}_e = 10$ dB.

Figure 4 compares the secrecy capacity for various antenna configurations, and the considered configurations are $(N_d, N_e) = (2, 3), (4, 3), (4, 4)$, and $(4, 5)$. For all cases, the secrecy capacity increases with the main-channel SNR. Increasing the number of destination antennas improves the secrecy performance by providing additional diversity, whereas increasing the eavesdropper antennas degrades it. Among the

considered configurations, $(N_d, N_e) = (4, 3)$ achieves the highest secrecy capacity, while $(N_d, N_e) = (2, 3)$ provides the lowest. This indicates that employing more antennas at the destination while limiting the eavesdropper's antenna diversity can significantly improve the overall secrecy performance.

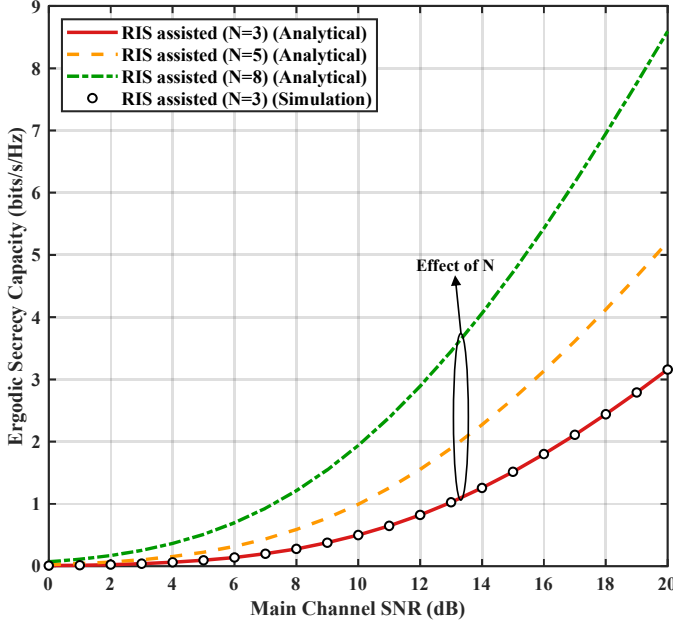


Fig. 5. Effect of N on ESC under MRC for $N_d = 2$, $N_e = 5$, and $\bar{\gamma}_e = 10$ dB.

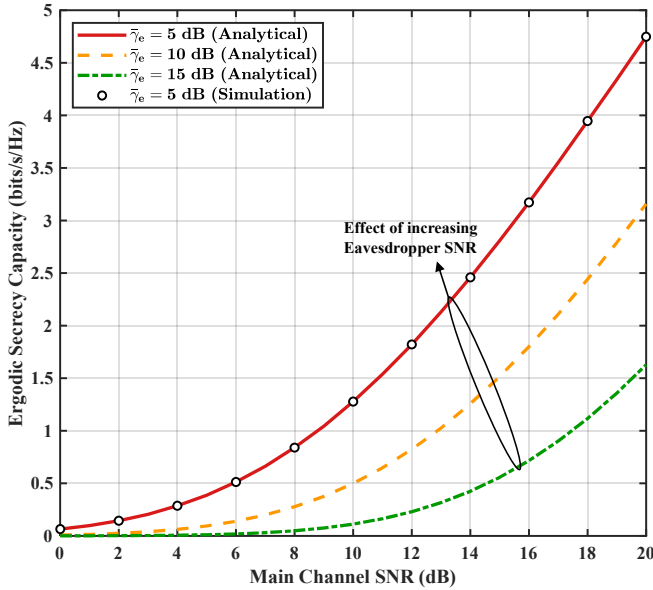


Fig. 6. Impact of $\bar{\gamma}_e$ on ESC under MRC for $N_d = 2$, $N_e = 5$, and $N = 3$.

Figure 5 highlights how the number of RIS reflecting elements can affect the achievable secrecy capacity when MRC is considered. The analytical curves are obtained for $N = 3$, $N = 5$, and $N = 8$, together with simulation points for the

$N = 3$ case. As the main-channel SNR is increased, a clear separation between the curves can be observed, and it becomes more pronounced at higher SNR values. The configuration with $N = 8$ provides the largest secrecy capacity, followed by $N = 5$, whereas $N = 3$ remains the lowest throughout the considered range. This behavior suggests that the use of a larger RIS leads to further improvements in the secrecy performance of the system.

Figure 6 demonstrates how the eavesdropper's average SNR affects the secrecy capacity when MRC is employed. Three different eavesdropper channel conditions are examined at $\bar{\gamma}_e = 5, 10$, and 15 dB. The red curve is higher than the other two over the entire range of the SNR in the main channel, showing that the system ensures a higher level of secrecy for a weaker eavesdropper channel. When $\bar{\gamma}_e$ is larger, the curves move downwards, and the loss in secrecy capacity becomes more noticeable at higher values of the main-channel SNR. This reflects the adverse effect of an improved eavesdropper reception capability on secure communication.

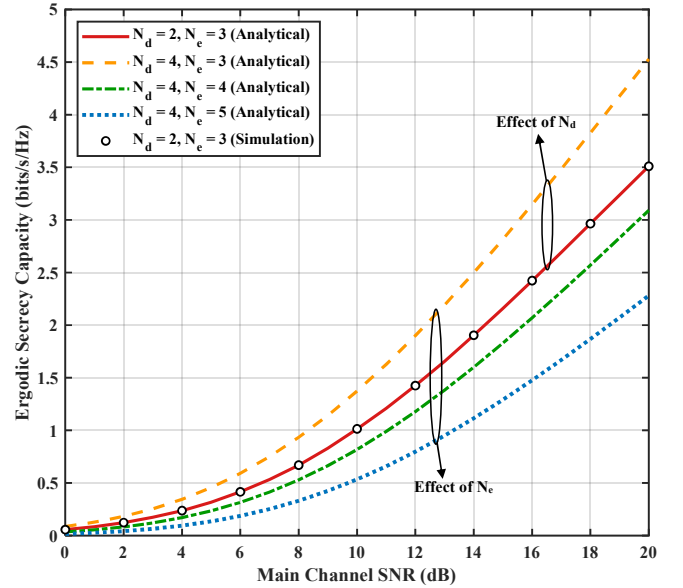


Fig. 7. Variation of ESC with respect to N_d and N_e under MRC for $N = 3$ and $\bar{\gamma}_e = 10$ dB.

Figure 7 compares the secrecy capacity obtained from different antenna arrangements at the destination and eavesdropper sides for MRC. The considered settings are $(N_d, N_e) = (2, 3)$, $(4, 3)$, $(4, 4)$, and $(4, 5)$. The curves reveal that providing additional antennas at the destination produces a clear gain in secrecy capacity, as reflected by the higher position of the $(4, 3)$ curve compared with $(2, 3)$. By contrast, when the number of eavesdropper antennas goes up from $N_e = 3$ to $N_e = 4$ and $N_e = 5$, the achievable secrecy capacity decreases. The differences are more pronounced as the SNR in the main channel becomes higher.

Based on the results obtained in Table II, it is found that for the considered range of N , N_d , N_e and $\bar{\gamma}_e$, the secrecy capacity of MRC is higher than that of SC for all

TABLE II
COMPARATIVE ESC PERFORMANCE OF SC AND MRC DIVERSITY
SCHEMES

Setting	0 dB		10 dB		20 dB	
	SC	MRC	SC	MRC	SC	MRC
$N = 3$	0.012	0.008	0.410	0.499	2.003	3.158
$N = 5$	0.021	0.024	0.515	0.994	2.194	5.208
$N = 8$	0.035	0.070	0.633	1.938	2.387	8.594
$\bar{\gamma}_e = 5$ dB	0.061	0.065	0.830	1.277	2.720	4.749
$\bar{\gamma}_e = 10$ dB	0.012	0.008	0.410	0.499	2.003	3.158
$\bar{\gamma}_e = 15$ dB	0.0011	0.0004	0.128	0.112	1.191	1.627
$(N_d, N_e) = (2, 3)$	0.023	0.057	0.512	1.014	2.188	3.509
$(N_d, N_e) = (4, 3)$	0.038	0.087	0.707	1.374	2.601	4.529
$(N_d, N_e) = (4, 4)$	0.027	0.035	0.632	0.818	2.490	3.091
$(N_d, N_e) = (4, 5)$	0.021	0.016	0.580	0.534	2.409	2.280

SC: Selection Combining; MRC: Maximal Ratio Combining.

the considered scenarios. When $N = 3$ is increased to 8 at 20 dB, the secrecy capacity of MRC is increased by 172%, while the secrecy capacity of SC is increased by only 19%, indicating that MRC gains significantly with increasing RIS elements. The results for the antenna side are in accordance with the diversity trend for both links, with the MRC secrecy capacity increasing by 29.1% when N_d is increased from 2 to 4 at 20 dB, and decreasing by 49.5% when N_e is increased from 3 to 5 at 20 dB, which is maintained at 0 dB and 10 dB as well. In comparison, the responses to the same antenna variations were only 18.9% and 7.4% for SC, respectively. The changes were much larger for MRC than for SC, indicating that MRC is more sensitive to these antenna variations when they occur at either extremity of the link. Similarly, for the average SNR of the eavesdropper $\bar{\gamma}_e$, increasing it from 5 dB to 15 dB degrades the MRC secrecy capacity by 66% while the secrecy capacity of SC is degraded by only 56%. The derived closed-form expressions are only finite double summations, so there is no truncation error when the values in Table II are calculated.

V. CONCLUSION

The analytical framework developed in this work shows that receiver-side diversity plays a significant role in the secrecy performance of RIS-assisted wireless networks. Exact closed-form expressions were derived for the ergodic secrecy capacity under both SC and MRC, providing a clear performance comparison and allowing the system designer to trade secrecy performance for receiver hardware complexity. The close agreement between the analytical and Monte Carlo results validates the derived expressions for practical use. The results also indicate that a stronger eavesdropper can be countered more effectively by expanding the RIS either by increasing the number of RIS reflecting elements or by adding destination antennas. This highlights RIS as an alternative approach to improving wireless security with receiver diversity. In future work, this framework can be extended to multi-eavesdropper scenarios, imperfect channel state information at the RIS, and joint optimization of the RIS phase configuration with the diversity combining strategy.

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